RIDOPANDI SINAGA

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience developing end-to-end AI solutions, spanning machine learning models and LLM-based applications. Experienced in Python, scikit-learn, and TensorFlow for data preparation, model development, and evaluation, alongside LangChain and LangGraph for building agentic RAG systems. Combines AI development with API and database integration to deliver reliable solutions for real-world business needs.

WORK EXPERIENCE

AI Developer - Freelance

Optik 150 K *Aug 2025 - Oct 2025*

- **AI Customer Service & Agentic RAG:** Built an end-to-end AI customer service agent using LangGraph and RAG to support product discovery, recommendations, price calculation, order creation, and payment assistance.
- **Human-in-the-Loop & Operations Dashboard:** Designed an operations dashboard for staff to monitor conversations, approve sensitive actions, take over customer chats, and return control to the AI while preserving conversational context.
- **Payment Reliability & AI Observability:** Integrated Midtrans QRIS with webhooks, polling, persistent state, and idempotency; implemented Langfuse to monitor LLM traces, token usage, costs, latency, tool execution, and RAG quality.

AI Developer - Contract

PT Riset Perkebunan Nusantara *Feb 2025 - May 2025*

- **OCR & Knowledge Engineering:** Built an automated OCR and document processing pipeline for **100+** unstructured commercial books, converting heterogeneous content into a structured, searchable, and traceable AI knowledge base.
- **RAG Chain Development:** Built RAG chains from scratch with LangChain and Qdrant, covering document parsing, chunking, embedding, vector indexing, retrieval, context construction, and LLM integration for document-grounded answers.
- **Retrieval Optimization & Reliability:** Refined chunking, metadata, retrieval settings, prompts, and chain orchestration; evaluated retrieved context and LLM responses to improve relevance, multi-document search quality, and system reliability.

AI Engineer Intern

Indonesian Oil Palm Research Institute (IOPRI) *Sep 2024 - Dec 2024*

- Improved chatbot Context Recall and Answer Relevancy by **10+%** through Neo4j knowledge graph integration and refinement of vector and GraphRAG retrieval across internal product information.
- Finetuned and evaluated a lightweight language model for structured Cypher query generation to support knowledge graph-based information retrieval.
- Evaluated retrieval and response quality using RAGAS, identifying failure cases and supporting iterative improvements to system accuracy and reliability.

EDUCATION & DEVELOPMENT

Universitas Sumatera Utara *2021 - 2026*

Bachelor of Computer Science, majoring in Information Technology | GPA: 3.92/4.00

Thesis: Implemented and evaluated agentic behavior in LLM-powered RAG pipelines by comparing traditional RAG with single-agent and multi-agent approaches using DeepSeek-V3. Evaluated retrieval and response quality using RAGAS and Wilcoxon signed-rank tests, increasing Context Recall from **73.60%** to **85.60%** with statistically significant improvement (**$p < 0.001$**) and achieving **92.16%** Answer Relevancy, with the strongest gains observed in multi-document retrieval scenarios.

Bangkit Academy led by Google, GoTo, and Traveloka *Feb 2024 - Jul 2024*

Selected as one of **4,650** participants from **55,000** applicants nationwide (approximately **8.5%** acceptance rate) for Bangkit Academy 2024 Batch 1. Completed the Machine Learning Learning Path through an industry-aligned intensive program covering Python-based data analysis, IT automation, machine learning, TensorFlow model development, and deployment.

ORGANIZATION EXPERIENCE

Google Developer Student Clubs (GDSC) USU - **Staff of Technical Division** *Sep 2023 - Jul 2024*

- Organized subject matter to design an interactive curriculum for the GDSC USU period, resulting in increased student satisfaction and a **22%** decrease in low-interest rates.
- Established connections with experienced speakers and secured their participation in workshops and events, resulting in a **35%+** increase in workshop registrations and **15%+** growth in online RSVPs.

AWARD

3rd Place Winner - Fordigi Hackathon Challenge 2023**, City Chapter Medan**

Placed **third** in the Medan chapter of a nationwide hackathon across **ten** Indonesian cities by co-developing Parentify, an AI-powered parenting education prototype addressing local health and hygiene challenges.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Data Processing & Cleaning, Exploratory Data Analysis
Software, Systems & Database: FastAPI, REST APIs, MySQL, Docker, Git, Model Deployment, Qdrant, Neo4j
Machine Learning & AI: Machine Learning, scikit-learn, TensorFlow, NLP, Information Retrieval, RAG, GraphRAG, Fine-tuning, LLM/Model Evaluation, Statistical Testing, RAGAS, LangChain, LangGraph, ReAct Agent, Langfuse
Certifications: Machine Learning Specialization · Process Data from Dirty to Clean · TensorFlow Developer Professional Certificate